Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/08/2016
PI Name:	Oates, Chris Ph.D.		
Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	11/16/2015	**End Date:**	11/30/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2018 per M. Lee/NASA HQ (Ed., 2/16/16)		
Key Personnel Changes/Previous PI:	EDITOR'S NOTE (December 2015): PI Chris Oates took over the project from Thomas O'Brian on 11/16/2015.		
COI Name (Institution):	Jefferts, Steven Ph.D. (National Institute of Standards & Technology)		
Grant/Contract No.:	NNH12AT81I		
Performance Goal No.:			
Performance Goal Text:			

EDITOR'S NOTE (December 2015): PI Chris Oates took over the project from Thomas O'Brian on 11/16/2015.
To support primary Atomic Clock Ensemble in Space (ACES) mission objectives, the National Institute of Standards and Technology (NIST) proposes to:

1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA (formerly known as the Joint Institute for Laboratory Astrophysics); and to support general ACES science goals.

2. Participate as fully as possible in coordinated European Space Agency (ESA) data analysis with formal ESA Data

Task Description:	Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources. As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental physics based on ACES data to include at least the following: • Global time and frequency transfer with uncertainty as low as 10-17. • Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10-17 or better. • Test of gravitation redshift at 2 x 10-6 uncertainty or better. • Search for variation in fundamental constants, such as fine structure constant at level of 10-18 per year or better. All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list): • World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10-16 or better. • Two mercury ion optical clocks with current fractional frequency uncertainty of approximately 1 x 10-17 and continually improving. • Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 8 x 10-18 and continually improving. • Two ytterbium lattice optical clocks with uncertainties better than 1 x 10-16 and rapidly improving. • Two strontium lattice optical clocks with uncertainties approximately 3 x 10-17 and rapidly improving. (At JILA. A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons.) • A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons. • A world-class time scale comprising seven hydrogen masers, with eight more masers to be added over the next three years. NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications. In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment. We propose the following revised program plan outline: Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction. Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure. Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time. Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time. Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major benefits and impacts of this research program can be grouped into three main areas: 1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. 2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory. 3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels

10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.

Task Progress:

In the most recent reporting period (November 2015 through November 2016) work has focused on two main tasks:
1. Preparation of the NIST site for the Microwave Link Ground Terminal (MWL GT) including progress on the NIST microwave and optical clocks.

2. Designing simulated data for testing the NIST data processing software in preparation for analyzing real data transmitted from the clocks onboard the ISS including relativistic effects and time delays.

Progress on each task is described below.

1. Preparation of the NIST site for the Microwave Link Ground Terminal (MWL GT) including progress on the NIST microwave and optical clocks

The Microwave Link Ground Terminal (MWL GT) system will be provided by ESA to participating ground station partners across the world, including NIST in Boulder, Colorado. The MWL GT systems are still under development through ESA sponsorship. NIST has been closely coordinating with ESA contacts on details of the MWL GT hardware, information technology and infrastructure requirements.

During the reporting period NIST has received the approved Letter of Agreement about ACES MWL GT installation at NIST, relating to the participation by NIST in the ACES mission. Specifications for the signals that NIST will supply to the ACES microwave link have been discussed by the NIST ACES group and plans are under way to provide these signals from the clock facility in NIST Precision Measurement Laboratory (Building 81). NIST is still awaiting the Ground Terminal hardware from ESA.

NIST continues to perform research on and improve the performance of its several optical and microwave frequency standards that will be crucial to international time and frequency comparisons and ACES-mediated fundamental research: two single-aluminum ion optical standards, one single-mercury ion optical standard, two ytterbium lattice optical standards, two strontium lattice optical standards, two cesium microwave primary standards, and multiple other frequency standards under development including calcium beam optical standards. In the past project year, the NIST ytterbium lattice optical clocks have reduced their fractional frequency uncertainty to below 2 x 10-18, a level they can now reach with only about one hour of averaging time. This ultimate stability enables the ytterbium lattice clocks to be exceptionally precise standards for fundamental physics research. This remarkable stability dramatically advances the goals of ACES time and frequency transfer given the relatively short visibility of the ISS to any MWL GT.

2. Designing and testing the NIST data processing software in preparation for analyzing real data transmitted from the clocks onboard the ISS.

A critical aspect of the ACES mission is accurate transfer of the time and frequency data between the International Space Station (ISS) and the ground stations. In particular, precise determination of the ISS orbit is essential to optimize the time/frequency transfer. Thus, during the reporting period, NIST's efforts in support of the ACES goals have principally been focused on the best methods for orbit determination through data analysis techniques. What makes this challenge particularly complex is the fact that data from the ACES equipment on the ISS will come not only from GPS receivers that estimate position and velocity, but from two-way ranging between ground-based monitors and the ISS, and possibly Doppler measurements of frequencies transmitted from the orbiting ACES oscillator. Finally, these methods for orbit determination ideally will be sufficiently accurate to test General Relativity with the ACES clock ensemble.

We have been working with the simulated ACES data supplied by the CADMOS (Center for Development of Microgravity Applications and Space Operations) software group in Paris. However, we have found problems with the provided data sets. In particular, we see large excursions in the z-component of angular moment of the ISS, large excursions in the energy per unit mass of the ISS, in spite of the fact that the orbit of the ISS in the simulation is stipulated to be Keplerian. As a result, we have been developing our own orbit simulations. We note the simulated data provided by ACES has still been useful because it shows what types of data will be available from the ACES package and the microwave link. Our software development so far can provide the following: Earth-Centered Inertial (ECI) coordinates and velocities of the ISS, of the NIST ground station, of the JPL (Jet Propulsion Laboratory) ground station, and ranging and propagation delays of one- and two-way signals between NIST and the ISS. A significant number of relativistic effects have been included, such as the Shapiro time delay (a few picoseconds), longitudinal and transverse Doppler shifts, bending of signal path as the signal follows a null geodesic, relative velocities of ACES and the NIST receiver. Extension of the simulation to similar signal exchanges between JPL and the ISS should be straightforward; this is of importance when the ISS is in common view of NIST and JPL.

We also emphasize that simulated ranging data from ISS to NIST location will have to be corrected for various delays. Ionospheric and tropospheric corrections are of the order of few meters and they have to be modeled and accounted for. A first take on tropospheric corrections has been implemented by adapting the routines used in precise point positioning (PPP) services supported by Natural Resources Canada (NRCan). We have implemented four different troposphere models and were able to set the uncertainty in estimation to be under 50 cm, larger than the 20 cm uncertainty set by the ACES mission requirements. As a result, we have begun to implement more sophisticated troposphere models that use actual meteorological data, such as the Vienna Mapping Function. Additionally, we are implementing the P3 ionospheric delay using higher order terms (the current model uses second order terms) involving dual carrier frequencies to obtain more accurate ionosphere-free pseudo-ranges.

Another critical aspect for the simulations will be to include the proper time elapsed on an ideal clock in a Keplerian orbit, geometric range between the ACES package and a NIST-based microwave link (MWL), time delay for signals passing from the moving ACES transmitter to the moving NIST receiver, and vice versa. We have included some effects that may be small for ACES such as the Shapiro time delay due to Earth's gravitational field, and the change of direction of the signals in Earth's gravitational field due to gravitational deflection of light. We have also included perturbing effects due to Earth's quadrupole moment in models of the elapsed proper time since these effects will be of order 10-13 or larger.

	Analysis of GPS data from the SIGI2 receiver on the ISS showed that the noise on position determination is mostly white. Accordingly, we have inserted white noise into our model orbit simulations; the simulations include relativistic effects. We performed a thorough quantitative analyses of ISS orbitography by comparing actual data for Dec. 10, 2010 with the simulated orbit that was determined by using mock ranging data for the ISS passage over NIST location. We found that the simulated data had a sinusoidal fluctuation with half the period of the actual data and the fluctuation in the z-comp of angular momentum 8500 times higher than is actually observed. We have eliminated quadrupole and higher order terms as the reason for this anomaly, and almost any other physical reasoning that might result in such behavior. Deviations from expected behavior of this type provided the motivation for us to develop our own simulated orbit data (described above) to better represent realistic orbit behavior.
Bibliography Type:	Description: (Last Updated: 04/05/2023)
Articles in Peer-reviewed Journals	Jefferts SJ, Heavner TP, Barlow SE, Ashby N. "Measurement of the microwave lensing shift in NIST-F1 and NIST-F2." Journal of Physics:Conference Series. 2016;723:1-7. (8th Symposium on Frequency Standards and Metrology 2015, 12–16 October 2015, Potsdam, Germany.) http://dx.doi.org/10.1088/1742-6596/723/1/012005 , Jul-2016